Semipply

A PRODUCT OF LEGALFORENSICS.AI

When the Chip Goes in a Car

The automotive contract terms that generic review misses

Qualification, PCN, longevity & the OEM squeeze · a Semipply Research Reader

SEMIPPLY RESEARCH

Semiconductor Supply Intelligence

The general field guide covers the twelve dimensions of risk on any semiconductor contract — take-or-pay, single-source supply, one-sided liability caps, the change-of-control exposure nobody signs. All of them apply when the chip goes in a car. But automotive adds its own grammar, and it is the part that surprises a buyer who treats a chip contract like any other purchase agreement.

Three things make automotive different: the parts are **safety-critical**, the programs run for **fifteen years or more**, and every part must pass a **qualification cycle measured in months** before it can ship. Together they turn ordinary contract terms into multi-year locks — and they create a handful of clauses that generic contract review, and even a general semiconductor read, will miss. This is a companion to the field guide, on what changes in a car.

Qualification lock-in — AEC-Q100, IATF 16949, PPAP

Before an automotive chip goes into production it clears three gates: **AEC-Q100** (the device passes automotive reliability stress testing), **IATF 16949** (the supplier's site runs an automotive quality system), and **PPAP** (a Production Part Approval Process sign-off that the specific part, from that specific line, is production-ready). That process takes months and real money.

The idiosyncrasy. Because qualification is expensive and slow, a qualified part is a *locked-in* part — switching sources means re-running the gates. Worse, the supplier can force that on you: a **fab move, a process change, or a site transfer can trigger re-qualification and a fresh PPAP.** The contract decides who bears that cost and that time.

Who carries it. Usually the buyer — the default is that a supplier can change its process and the burden of re-qualifying lands downstream. **Advice:** pin down, in writing, who pays for and schedules re-qualification on a *supplier-initiated* process or fab change. A clause barring unilateral process changes without re-qualification at the supplier's cost is worth far more than it reads.

PCN flow-down — and the notice gap

A **Product Change Notification (PCN)** is the advance warning a supplier owes before it changes or discontinues a part. In consumer electronics it is a courtesy. In automotive it is load-bearing, because of a gap most buyers never measure.

The idiosyncrasy. The PCN **notice window** in your supply contract is often *shorter* than the **re-qualification timeline** the change forces on you — an AEC-Q100 re-qual runs roughly six to eighteen months depending on the device. So you can receive perfectly compliant notice and still be caught short, with no qualified part when the change lands.

Who carries it. The buyer, sitting in the gap between the notice window and the re-qual clock. **Advice:** measure the two against each other. If your PCN notice is shorter than your device's re-qualification time, that difference *is* your exposure — negotiate the window up, or secure a bridge-supply obligation to cover it.

Longevity, EOL, and the last-time-buy

Cars are built for years and serviced for a decade or more after that. A chip's commercial life is shorter — the supplier would rather move you to the next part.

The idiosyncrasy. The **end-of-life (EOL)** and **last-time-buy (LTB)** terms decide whether you can secure supply across your whole program and service tail, and on what notice. Without them, a supplier can discontinue a part on its schedule, not yours, and leave you unable to build or repair a platform still on the road.

Who carries it. The buyer left holding an un-serviceable program. **Advice:** get a **longevity commitment matched to your program life**, plus a **last-time-buy window long enough to bridge to a redesign** — and confirm the LTB quantity math actually covers production *and* the service tail, not just the next model year.

The OEM directed-buy squeeze

Since the 2021–22 shortage that idled auto plants worldwide, carmakers have gone **direct** to chipmakers — negotiating supply and, increasingly, **directing the Tier-1 on whom to buy from** (GM with Wolfspeed and Qualcomm, Ford with GlobalFoundries, among others).

The idiosyncrasy. The Tier-1 keeps the **purchasing obligation and the risk** but loses the **leverage** — it is committed to a source it did not choose and cannot freely renegotiate, while the OEM holds the relationship above it. It becomes the meat in the sandwich: bound to the supplier's terms below and the OEM's direction above.

Who carries it. The Tier-1, squeezed on both sides. **Advice:** separate what is **OE-directed** from what you actually control — your negotiating leverage lives in the non-directed parts. Where a part is directed, push to **flow the OEM's commitments back-to-back** onto the supplier, so you are not carrying a take-or-pay to a source the OEM chose without the OEM standing behind the volume.

Where the general dimensions bite harder

The field guide's twelve still apply — they just cut deeper in a car:

- **Take-or-pay** against a program that can be cancelled or slowed: you are committed to volume the vehicle line may never draw.
- **Single-source** on a safety-critical part with no qualified alternative — the exact geometry that idled plants in 2021, and the reason the Nexperia reading matters.

- **Change of control** on a supplier whose ultimate owner you cannot see — a cheap part, one plant, an owner your screens miss.
- **Force majeure and allocation** in a shortage: you are one of many customers, and the allocation rule is written by the party who is short.

The through-line

The general field guide tells you *who carries the risk* on any chip contract. Automotive multiplies that risk three ways — the **qualification lock** that makes every part hard to leave, the **service-life horizon** that outlasts the part, and the **OEM above you** that holds the leverage while you hold the obligation. Read those before you sign a multi-year program, not after a line goes down.

A Semipply Research automotive reading, and a companion to The Idiosyncrasies of Semiconductor Contracts. *AEC-Q100, IATF 16949, PPAP, PCN, and last-time-buy are standard automotive-industry practices; this guide describes how they shape contract risk. Companies named are cited from the public record. Automotive component-supply agreements are largely not publicly filed, so this is a domain guide grounded in industry practice and public reporting — not a corpus benchmark. Not legal advice.*